

Neutral Tone Mapping for PBR Color Accuracy

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ABSTRACT

On e-commerce websites, interactive 3D product models are side-by-side with sRGB product photos that have been carefully color graded in post to achieve the desired marketing look. For color consistency, e-commerce production requires a tone mapper that faithfully reproduces 3D model material colors on the screen under neutral (grayscale) lighting to match the photos. This allows artists to build 3D models using marking-approved sRGB color swatches without needing to later tweak the material values to make the output align to existing product images.

A neutral tone mapper has been developed at the Khronos 3D Commerce working group precisely to address this need. The goal is to standardize it and make it an available option across authoring tools and renderers as an improved alternative to disabling tone mapping entirely when no "look" is desired.

KEYWORDS

tone mapping, color grading, physically-based rendering, e-commerce

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1 INTRODUCTION

The difficulty with compressing a nearly infinite HDR range down to sRGB is the necessary loss of information. For the highlights to be discernable, the neutral contrast must be reduced, so a paper-white object must appear dimmer than 255 under diffuse lighting. For the highlights to desaturate appropriately and smoothly, the highest saturation colors of sRGB become unreachable under any lighting.

The primary problem with ACES tone mapping reported by e-commerce users is their significant loss of saturation. Because common Physically-Based Rendering (PBR) pipelines specify material and lighting colors in sRGB (e.g. glTF files), the linear output values are always within the Rec.709 gamut. Figure 1 shows the reachable set of ACES tone mapping under this restriction, compared to our PBR Neutral tone mapper. Note that canary yellow,

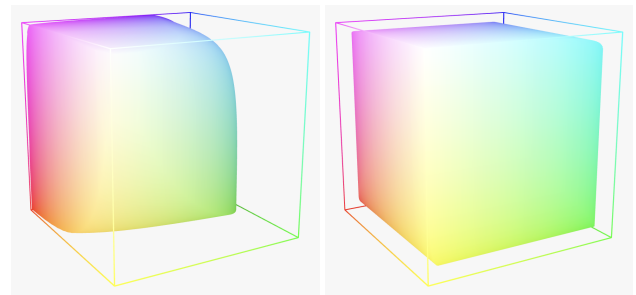


Figure 1: Comparison of the reachable set of sRGB of ACES (left) vs PBR Neutral (right) tone mappers for Rec.709 input. This is plotted in linear space, not gamma-encoded.

bright greens and blues are all impossible to output to the screen under ACES, as well as other filmic tone mappers such as AgX.

When no filmic "look" is desired sometimes tone mapping is disabled entirely, which creates a linear output, clamped at 1. This results in blown-out highlights and significant hue skews due to the channels clamping individually. See Figure 2 for a comparison of these tone mappers and note the hue skews and desaturation.

2 PBR NEUTRAL TONE MAPPER

The PBR Neutral tone mapper is designed to be simple to implement, fast to run, and faithfully reproduce color as much as possible while eliminating HDR artifacts around highlights. It is intended to be 1:1 for colors up to a certain maximum value, with the remainder used as headroom for the compressed highlights.

Even without tone mapping, PBR already desaturates colors, particularly dark dielectrics, due to the Fresnel effect, which adds 4% of the incident light color (for default IoR of 1.5). The PBR Neutral tone mapper corrects for this on average by shifting the 1:1 portion of the curve down by 0.04. This allows us to add a "toe", which helps the blacks retain better contrast. By using the minimum of R, G, and B, this toe is smooth, with derivatives that vanish at all the faces of the output sRGB cube.

The brightness, defined here as the maximum value of R, G, and B, is reduced to the 0-1 range by a scalar multiplier, thus preserving hue. At extreme brightness the color is desaturated, because perceptually that color is lost when it is sufficiently bright. Desaturation is based on the amount of brightness removed and asymptotes out much slower. In a sense we are replacing the lost brightness with desaturation, thus giving the brain an alternate perceptual cue, which smoothly encodes several orders of magnitude more brightness than is available in the output screen.

Simple functions with minimal parameters were used to match slopes and create asymptotes. Parameter values were chosen to

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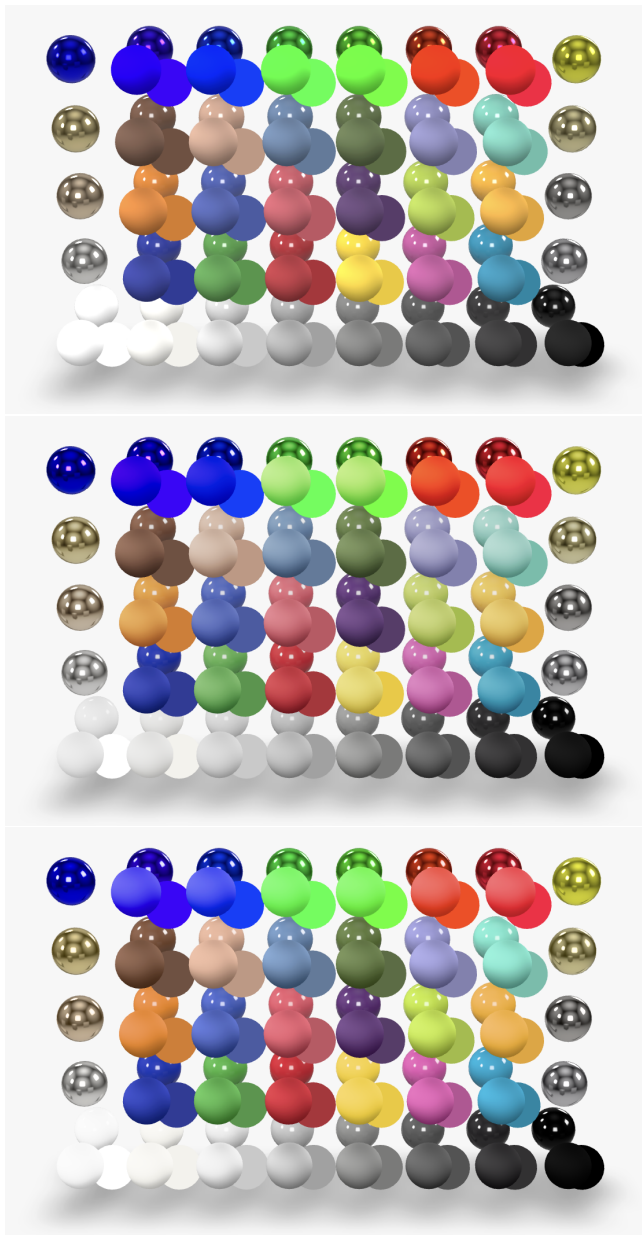


Figure 2: A 3D model inspired by a Macbeth chart, including rough, shiny, and reference spheres for each color, with metallic and saturated spheres along the top and sides. Comparison of no tone mapping (top), ACES (middle), and PBR Neutral (bottom) with the same grayscale studio lighting.

avoid compression artifacts in highlights while maximizing the output sRGB range. Material color values are exactly reproduced under analytic lighting for sRGB values between 80 and 231 (0.08 and 0.8 in linear).

The above curves are shown in Figure 3. Note that despite the piecewise definitions, the first derivatives are continuous everywhere and the function is analytically invertible. The complete

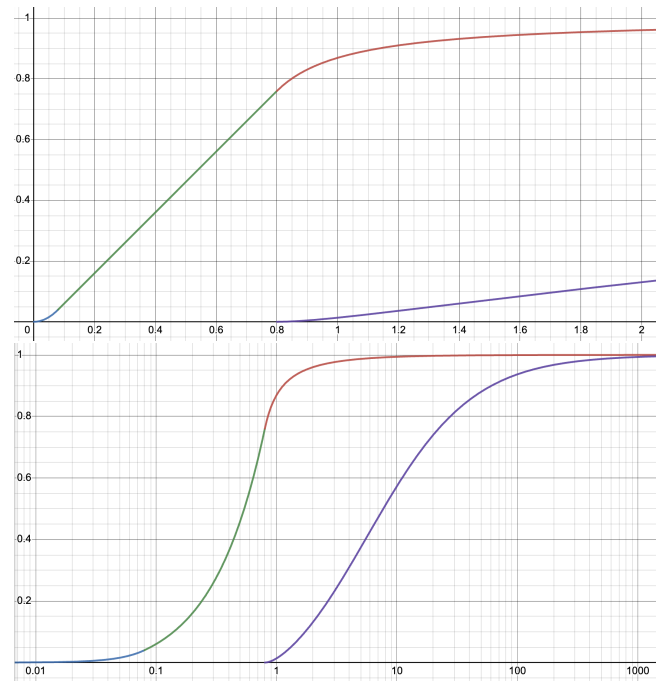


Figure 3: The PBR Neutral tone mapping function, which maps linear brightness to the $[0, 1]$ range (top: linear, bottom: log). The compression function is piecewise in three parts: blue for the toe, green for the 1:1 region, and red for the compression region. The desaturation curve is purple.

shader code is quite small, with only three divides and those only applied to colors over the 1:1 limit:

```
vec3 PBRNeutralToneMapping( vec3 color ) {
    const float start = 0.8 - 0.04;
    const float desat = 0.15;

    float x = min(color.r, min(color.g, color.b));
    float offset = x < 0.08 ? x - 6.25 * x * x : 0.04;
    color -= offset;

    float peak = max(color.r, max(color.g, color.b));
    if (peak < start) return color;

    const float d = 1. - start;
    float newPeak = 1. - d * d / (peak + d - start);
    color *= newPeak / peak;

    float g = 1. / (desat * (peak - newPeak) + 1.);
    return mix(newPeak * vec3(1, 1, 1), color, g);
}
```

You can find the specification of the Khronos PBR Neutral tone mapper at <https://github.com/KhronosGroup/ToneMapping>, along with a corresponding OpenColorIO profile for usage within existing workflows. Look for this view transform in Blender 4.2, as well as Three.js, Filament, PlayCanvas, Babylon.js, and more.